

Harmonized Banner Creation from Multimodal Design Assets

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ABSTRACT

Designing aesthetically pleasing single-page graphic designs ("banners") that appeal to the target recipients is non-trivial and requires considerable human effort. Designers often start by collecting assets like background images, inspirational banners, and relevant taglines/textual phrases, and iterate over variants of the designs by testing different combinations of these assets for their current task. To expand creative processes beyond professionals, it is crucial to accelerate this tedious process of creating design variants. To this end, we propose an algorithm that takes as input multiple image and text elements to generate several visually pleasing banner designs harmonized for text colors, font styles, and the placement of these multimodal assets. The generated layout accounts for background image saliency and uses this to select typographic and color variants yielding the harmonized outputs. We demonstrate the effectiveness of the proposed algorithm in generating aesthetic banners with the help of a crowdsourced experiment.

CCS CONCEPTS

• **Information systems** → **Multimedia content creation**; • **Computing methodologies** → **Image-based rendering**.

KEYWORDS

banner creation, single page graphic design, automated systems, style harmonization

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1 INTRODUCTION

Designing banners that are visually appealing to the target recipients is a non-trivial task and requires considerable human effort. Once created, the set of banners and their designs influence customer interest and play an important role in optimizing click-through rates. Unlike a generic image, a banner is an ensemble of an intent, a theme, a collection of images, affiliated text phrases (in a specific font style - typography, size, and color), and additional elements including logos, shapes, and patterns [22, 28]. In the field of visual design, designers collate inspirational designs, images and various relevant multimodal assets before coming up with a new design by combining a subset of these assets [25]. When designing a new banner, a lot of time goes towards repetitive tasks [28] like (i) choosing different combinations of image and text fragments, (ii) arriving at an appropriate layout, (iii) optimizing color & font style choices for different text elements so as to complement the background regions, and (iv) evaluating the visual effectiveness of the banner. The process of generating these variants is tedious and requires significant manual effort, calling for automation in creating harmonized design variants from the collected unstructured image and text content.

We propose a multi-stage pipeline to automate this design variant creation by generating content-aware layouts from background images. We identify salient and non-salient regions of the images to determine the position of textual elements that are to be placed on top of the background images. A genetic algorithm-based technique is used to optimally minimize the energy while accounting for three dimensions - visual saliency, alignment, and overlap. The generated layouts are then used to contextually select harmonized colors and font styles for the text, yielding a final design suggestion. Fig. 1 shows a set of banners generated from our pipeline depicting how the algorithm can be used to generate multiple design variants by utilizing the assets collected by the designer.

2 RELATED WORK

Works on **layout optimization** have explored generating layouts in content-aware or content-agnostic fashion. Content agnostic approaches generate wireframe-like layout structures optimizing the placement of specific classes of elements [1, 8, 15, 18–20]. These approaches do not consider the context of the elements or background image when optimizing the layout. In [27], O'Donovan et al. contribute an energy function devised from various design principles to optimally synthesize single-page graphic designs and



Figure 1: Banner design variants generated by the proposed system

further extend it to an interactive setup in [26]. Maheshwari et al. [23] extend the energy functions to transfer and adjust content to mimic the look of an example banner. While we take inspiration from these works, we note that they are also content-agnostic. Additionally, works such as Jyothi et al. [15], Li et al. [19] expect a label set of classes to which elements may belong and generate stochastic outputs without allowing for conditioning, hence unsuitable for generating layouts from a unique set of elements.

Recent works [18, 20, 31] allow the conditional generation of layouts, which is more aligned to our goal. Lee et al. [18] allow specification of a partial directed graph capturing spatial relations between elements for optimization. Li et al. [20] propose the Attribute-Conditioned LayoutGAN, capable of synthesizing layouts through geometric relation modeling among distinct types of 2D elements, conditioning it on the reading order of elements, expected size of a design asset, and aspect ratio. However, these constraints are also content-agnostic and do not extend to content-aware color and font harmonization – which are critical to banner generation. Zheng et al. [31] propose a generative model to synthesize magazine cover designs conditioned on constraints such as the category of magazines. Zhang et al. [30] propose an approach to detect and place a single text snippet on an image using the saliency information. While these works are content-aware, they operate on abstract constraints and does not extend to the needs of the banner design optimization that we are exploring. Further, both Li et al. [20] and Zheng et al. [31] focus on generating layouts by optimizing for geometric parameters while Zhang et al. [30] is limited to the placement of single text box on an image. All these works fail at multi-object placement on top of an image by capturing the saliency and context information of the background to create banner variants, which is our proposal in this work.

Another line of related work deals with **automating stylistic choices** such as color or font style selection. We define *harmonization* of graphic design as a broad concept ranging from recolorization of images, to automated cropping, to stylistic rendering of fonts. We specifically focus on harmonizing color and font choices with respect to the banner image. These choices determine if a banner is visually appealing or not. Rendering a dark colored text on an already dark background region will be unpleasant, thus making color harmonization a fundamental requirement to design aesthetic banners. Visual Font Pairing [14] aims to automatically identify the

pairs of fonts that can go together in a visually pleasing manner. FontMatcher [5] models the relation between images and its suitable fonts and generates descriptions to explain the choice of font by correlating various objects to the chosen font. Both these works focus on making harmonized choices when one of the element types is fixed. FontMatcher relies on the objects present in the image to generate a caption and use this to arrive at a font choice. The technique is therefore limited to images with recognizable objects and does not extend to abstract or natural image backgrounds. We therefore use a more generic framework that uses the entire image context for picking fonts that go well with the input image. In the color domain, works like [7, 17] extract color palettes from images using K-means clustering and grouping of colors while preserving diversity. Jahanian et al. [12] use color theories [11, 16] to choose appropriate colors for texts on a magazine cover.

We propose a process that can generate layouts for the placement of multiple design elements on the background image by considering the visual saliency of the image as well as alignment and overlap between pairs of elements. Prior work focused on either placement of a single element on non-salient regions or on generating an aligned layout between the different elements with no regard to saliency/context. Secondly, for harmonizing the final banner, we develop a two-step pipeline that can take the generated layout and harmonize the text elements in terms of color and font styles, based on the semantic information of the background region against which they are placed. Prior work in this area primarily relies on predefined templates and heuristics for style choices. Our key novelty is in proposing a new technique for creating banner designs that is more data-driven and adaptable to a variety of contexts, furnishing an end-to-end pipeline for generating banner variants that can readily be used to kickstart creative processes.

3 METHODOLOGY

Fig. 2 depicts our overall pipeline for generating harmonized banner designs. ResNet [10] representations extracted from the image provide contextual information that can be used to predict appropriate font styles that ‘go well’ with the image content. Additionally, extracted image saliency features [3] influence layout optimization, allowing identification of less-salient regions on which text may be placed. We use the predicted layout information along with text to arrive at a color choice that accounts for text visibility and contrast

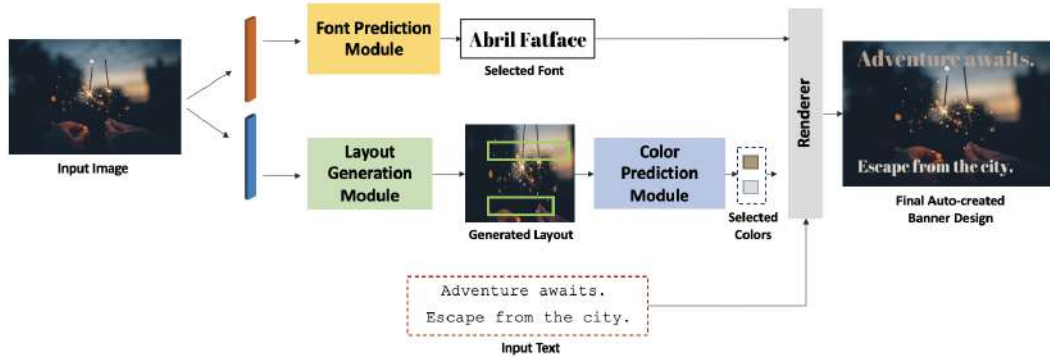


Figure 2: Auto-Harmonization pipeline to generate banner design variants

in the placed region. Finally, the chosen font style, color, and layout are used to render the final banner design suggestions.

Content-Aware Layout Generation: A major component of a coherent banner design is the visual-textual arrangement of its constituent elements. We consider automatic harmonization for a single page banner design, where there is a background image and several text elements for which the position and intra-text formatting need to be inferred. The arrangement of these elements is dependent on the background image. For example, we would choose not to place headings on salient parts of an image. We propose a genetic algorithm framework that optimizes places of multiple elements conditioned on the saliency of the background, while preserving traditional notions of alignment and overlap.

The algorithm takes as inputs the width and height of bounding boxes corresponding to each text element, wherein the number of text elements and their contents may be input by the user. We initialize a population of candidate layouts, define an energy function to be minimized, and construct a breeding mechanism that leads to optimal layouts being generated from the most viable candidates in each iteration. The layout of a banner design may be represented by specifying the center coordinates (x_c, y_c) for each element, treating the top left corner as the origin $(0, 0)$ and measuring along the width and height parameters w, h for that element. We treat this layout representation to be independent of the size of a background image. To do this, the center coordinates, and height and width of each element, are normalized to lie in the range $[0, 1]$ as follows: x_c & w are each expressed as a fraction of the width of the background image whereas y_c & h are each expressed as a fraction of the height of the background image. By specifying coordinates as fractions of the dimensions of a background image, the layout can be easily rendered on backgrounds of varying sizes. The expected input parameters w & h can be varied automatically to generate different layouts for the same banner elements. We now describe these stages in greater depth.

Initialization: Each candidate layout in the initial population is generated by selecting center coordinates for every element, uniformly sampling from the range of valid values for each coordinate. In determining the valid range for a coordinate, we keep in mind that the center (x_c, y_c) of an element of width w and height h should be positioned such that the edges of the element do not overflow the bounds of the background frame. In addition, we make sure to leave

a margin m_w from the left and right sides of the background image, and a margin m_h from the top and bottom sides, to avoid forcing text elements to the edges of the banner. Hence, each coordinate can take values in the range $\frac{w}{2} + m_w \leq x_c \leq 1 - \frac{w}{2} - m_w$ and $\frac{h}{2} + m_h \leq y_c \leq 1 - \frac{h}{2} - m_h$, where, $w, h, m_w, m_h, x_c, x_y \in [0, 1]$ as described earlier.

Fitness: We evaluate the fitness of a candidate layout by computing an energy function $\mathcal{E}$, defined as a weighted combination of three components – a visual saliency score $\mathcal{L}_{sal}$, an alignment score $\mathcal{L}_{al}$ and an overlap score $\mathcal{L}_{ov}$; given by $\mathcal{E} = w_{sal} \cdot \mathcal{L}_{sal} + w_{al} \cdot \mathcal{L}_{al} + w_{ov} \cdot \mathcal{L}_{ov}$. Fig. 3 pictorially depicts the intuition behind each of these.

Visual Saliency Score ($\mathcal{L}_{sal}$) quantifies the salient portion of the background image that would be obscured by text in the current layout. We first extract a heatmap H representing the salient regions of the background image X using the model in [3]. For every pixel i in X , the corresponding output importance prediction in H is $P_i \in [0, 1]$, where larger values signify greater importance. As we focus on placing elements in the non-salient regions of X , we are interested in the inverse saliency map H' which assigns larger values to pixels of less importance: $H' = 1 - H$, where 1 is a 2D matrix consisting of all ones, of the same size as H . Given a candidate layout l , we also construct H'_l , which represents the inverse saliency map deduced from the specific layout in question. In this construction, we treat all pixel positions in the background image that would overlap with elements in the current layout as assumed to be of least importance by the algorithm (represented as 1), and all non-overlapping pixel positions to be most important (represented as 0). Hence, H'_l is constructed by applying a layout mask of 1's to an initial array of 0's in the shape of X . The resulting algorithm tends towards producing layouts that prioritize placement in non-salient regions of the background image. To drive the placement of elements to higher valued prediction regions in H' , we also penalize layouts that obscure salient regions of the image by computing the mean of the element-wise difference between H'_l and H' given by $\mathcal{L}_{sal} = \frac{1}{N \times M} \sum_{i=1}^N \sum_{j=1}^M \max(H'_{l(i,j)} - H'_{(i,j)}, 0)$, where, N and M are the height and width of the background image. We ignore negative outputs of the element-wise difference as they indicate the regions in the layout without elements that overlap with the non-salient regions of the background image and is an acceptable outcome.

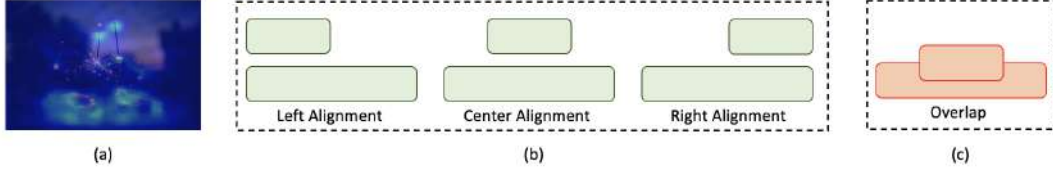


Figure 3: Proposed approach for layout generation optimizes for (a) preserving visual saliency, (b) maximizing intra-element alignment & (c) minimizing intra-element overlap

Misalignment Score ($\mathcal{L}_{al}$) encourages alignment along the left or right borders, or central axis, among the text elements within a banner, given by, $\mathcal{L}_{al} = \min(al_{center}, al_{left}, al_{right})$, where $al_{center} = \sum_{i=1}^n \sum_{j=i+1}^n abs(x_{c_i} - x_{c_j})$, $al_{left} = \sum_{i=1}^n \sum_{j=i+1}^n abs(x_{l_i} - x_{l_j})$ and $al_{right} = \sum_{i=1}^n \sum_{j=i+1}^n abs(x_{r_i} - x_{r_j})$. n refers to the number of elements in the banner, x_l & x_r are the left and right x coordinates for each element in the banner, calculated as: $x_l = x_c - \frac{w}{2}$ and $x_r = x_c + \frac{w}{2}$. We penalize differences in x -alignment, but allow the algorithm to orient equally for left, right or center alignments, thus providing freedom in placement within the visual saliency constraint. Since it is sufficient for elements to align along any of the 3 orientations, we take the least misalignment output across orientations as the penalty term.

Overlap Score ($\mathcal{L}_{ov}$) discourages elements from overlapping with each other and is given by $\mathcal{L}_{ov} = \sum_{i=1}^n \sum_{j=i+1}^n IoU(i, j)$ where, $IoU(i, j)$ is the standard *Intersection over Unionscore* between elements i and j of the banner.

Selection: All individual layouts in a population are evaluated using the fitness function $\mathcal{E}$ (we seek to minimize this value for the optimal banner). To select viable individuals that may proceed to the next iteration, we eliminate all individuals from the population that have an energy value higher than the mean for that population. Consequently, to maintain the size of the population, we perform a reproduction procedure on the individual layouts that remain.

Reproduction We generate new individuals to maintain the population size and proceed to the next iteration by modifying the viable individuals from the previous iteration – seeking to propagate successful characteristics. The reproduction operation given two viable layout individuals, for a single element of the banner, is given by, $(x'_c, y'_c) = \left(\frac{x_c^p + x_c^q}{2} + n_1, \frac{y_c^p + y_c^q}{2} + n_2 \right)$, where $n_1, n_2 \sim \mathcal{N}(0, 0.005)$. Here, (x'_c, y'_c) represent the center coordinates for the banner element of the new individual layout, (x_c^p, y_c^p) and (x_c^q, y_c^q) represent the center coordinates for the corresponding banner element of the p^{th} & q^{th} individuals respectively, where p and q are randomly chosen from between 0 and n_v (number of viable individuals remaining after elimination). Further, we apply a random mutation to each of the coordinate values for a given element by injecting noise sampled from a standard normal distribution of variance 0.05. The same operation is repeated for each banner element. In our implementation, we run 1000 iterations of the algorithm, using 10 individuals to form a population. We empirically fix the weights w_{sal} , w_{al} , & w_{ov} as 1000, 50, and 100, respectively, and margin values m_w and m_h as 0.05.

Harmonized Color Selection for Text Elements. On studying the design of banners, it is evident that the choice of color applied to any text element on a background image is dependent on the specific region on which it is placed. Leveraging works in color theory [9, 13, 24] on luminance and the contrast ratio between text and its background, we design an approach for inferring a suitable aesthetically pleasing color to be applied to the text placed on an image. We first identify the dominant color of the portion of the image as cropped using the predicted text bounding box dimensions from the predicted layout. We also extract the color palette of the entire image, represented as 5 (R,G,B) color bins. We then determine the color from this palette that has the greatest contrast ratio with our identified dominant color and present this as the predicted color choice. This ensures clear visibility, thus adhering to color theories. This is also harmonious with the rest of the image since this the chosen color is already present within it.

For identifying the dominant color in a cropped image, we apply the Modified Median Cut Quantization (MMCQ) [2] algorithm to quantize the pixels in the image region into 5 bins, $\mathbf{d} = MMCQ(X_{x_1, y_1}^{x_2, y_2})$ where, $\mathbf{d}$ is a vector of length 5 (by design) containing the (R, G, B) values of the identified palette bins, sorted in decreasing order of the number of pixels assigned to each as an outcome of the quantization algorithm. X is the background image while (x_1, y_1) and (x_2, y_2) are the top-left and bottom right coordinates of the part of the image for which we want to identify the dominant color.

Following quantization, we identify the bin having the largest number of pixels as the bin corresponding to the dominant color (R_D, G_D, B_D). In a similar fashion, we quantize the pixels in the entire background image into 5 (R, G, B) color bins as $\mathbf{p} = MMCQ(X)$. Following the guidelines in Caldwell et al. [4], we then determine the color from the palette $\mathbf{p}$ that exhibits the greatest contrast ratio with the dominant color (R_D, G_D, B_D) which we present as the harmonized color choice for the text element. Our auto-harmonization methodology also supports an easy transition from the color identification pipeline described here to techniques which involve a more nuanced understanding of color theory and design.

Harmonized Font Style & Typography Determination. The choice of an appropriate font for a background image is inherently influenced by image content and style. Concretely enumerating these factors as a set of heuristics is both challenging and generalizes poorly. We adopt a data-driven strategy by modelling font recommendation as a classification task – given an input image, we obtain a probability distribution over all font classes in the database and suggest the fonts with greater probability scores to "pair well" with the image. To prioritize the context of the background

image in font choice, we employ a pretrained ResNet [10] trained on ImageNet [6] dataset to obtain image representations. These representations are passed through a one-layer feed-forward neural network, whose output is a 58 length vector with a score for each of the 58 font classes in the dataset, and optimized on the Cross Entropy Loss.

Following identification of the font class, we employ a binary search based approach to jointly infer the font size and typography for each text element, given its predicted bounding box dimensions. This algorithm takes as input the height and width of the predicted bounding box for the text element, along with the font name and text contents. In order to determine the appropriate text indentation, we start by constraining the current number of characters in a single line to be no less than a chosen minimum and render the text in the predicted font style for different indentations and font sizes. This allows us to choose a setting in which the area occupied by the rendered text reasonably covers the bounding box region.

4 METRIC-BASED EVALUATION

Fig. 2 depicts the stylistic choices made by our pipeline in generating a banner design, showing an easy dissection of the final rendered design into its constituent stylistic preferences. To evaluate the performance of the proposed approach for content-aware layout generation, we run the same pipeline with all the modules intact but use the Attribute Conditioned LayoutGAN [20] as the underlying layout generation module, as a baseline against which to compare. Li et al. [20] propose the Attribute Conditioned LayoutGAN for conditional layout generation and demonstrate its use in automating advertisement poster and banner structural design. It takes the geometric parameters of various assets that are to be placed on a blank canvas and optimizes the size and positions to generate a suitable layout. The major limitation of such a layout is its content-agnostic nature. The LayoutGAN relies entirely on geometric parameters and incorporating content information in the training process is not straightforward. The parallels and distinctions between the LayoutGAN and our approach make it a natural baseline for evaluation.

Fig. 4 presents a few generated banner designs to compare layout generation using our approach against the Attribute Conditioned LayoutGAN. We observe that, across all these examples, the LayoutGAN appears to place text segments in the top and central regions of the images and aligns all text elements centrally, irrespective of the choice of background image. On the other hand, banners from our saliency-aware approach allow text elements to shift flexibly to accommodate the saliency of the content on which they are placed. To compare the performance of the generations quantitatively, we evaluate the generated layouts on two grounds: (i) Intersection over Union (IoU) between the predicted layouts and the ground truth originals, to gauge the performance of each algorithm against human choice, and (ii) Visual Saliency scores (defined as $\mathcal{L}_{sal}$ in our energy function for layouting), to quantify the portion of salient information in the background image that is obscured in the generated layouts. Ideally, we desire that the generated layouts are closer to human layout choices (higher IoU scores) and do not obscure important content of the background image (lower Visual Saliency scores). The **IoU and Saliency scores** for our approach

were **0.1049** & **0.0195** respectively which were significantly better than the LayoutGANS where the scores were **0.0290** & **0.0384**.

To train and test the font selection module, 29,415 banners from our proprietary dataset were used to obtain the corresponding background images (with no text element) and ground truth font labels. Note that each banner image can have multiple textual segments with different font styles. We choose the font of the largest text element in the banner as the ground truth value for that sample. There are a total of 58 unique font classes in this dataset. We split the 29,415 image-font pairs in ratio 8 : 1 : 1 to form train, validation, and test sets respectively. We train the model for 100 epochs, choosing the learning rate as 0.0001, and batch size as 64. An accuracy score of 24.55% is achieved on the test set using this model. Even though the accuracy metric used here is strict, (since in literature, generally top-3 or top-5 most probable choices are taken to evaluate the F-scores and accuracies [29]) we can still observe a clear advantage of having a context aware font selection model as compared to a random font selection which would give be equivalent of selecting 1 out of 58 classes randomly giving a rough accuracy score of 1.72%.

5 HUMAN EVALUATION

Design aesthetics are highly subjective and influenced by the perception of individuals. There is no comprehensive way to evaluate the overall pipeline through a metric-based method. We therefore resort to a large scale crowdsourced experiment to determine the efficacy of our approach in design variant generation.

To formally evaluate the proposed pipeline for auto-harmonizing multi-modal assets (images + text elements) to create banner variants, we generated a set of banner designs in which the layout for text placement was determined independently via both (a) the Genetic Algorithm (Our approach) and (b) the Attribute-Conditioned LayoutGAN [20] (our baseline). To generate these banners, we first selected a random set of 50 banners created by designers. We then used background images and size of text elements from these banners in order to generate new layouts using both the aforementioned approaches. The color and font selection module remained common for both techniques. Thus, two sets of banners were created - one with Genetic Algorithm (GA) technique (set A) and the other with the LayoutGAN (set B). We deployed a human experiment to compare banners from set A & B.

We asked Amazon Mechanical Turk (MTurk) annotators to observe the two banner images (one from set A and the other from set B) and answer a series of questions. The designs were evaluated along three dimensions: (a) the overall "visual appeal" of the design, (b) effectiveness of the generated layout, and (c) aesthetic appeal of the color choices. The annotators answered the questions on a 5-point Likert Scale [21] with 1 as the lowest score and 5 as the highest. High scores are desirable for all 3 dimensions. Each image pair was evaluated by 5 different AMT workers. To maintain the quality of the annotations, workers were restricted to those that have a 95% or higher approval rate and a minimum of 100 approved annotations. Each task took less than 1 minute as per our pilot surveys. It is evident from Table 1 that the banner designs generated using our approach received significantly better ratings than



Figure 4: Banners Generated using LayoutGAN (a & c) vs using our approach (b & d)

Framework	Visual Appeal	Layout		Color Harmonization
		Saliency	Alignment	
LayoutGAN based approach [20]	3.17 $\pm$ 0.71	2.92 $\pm$ 0.86	3.59 $\pm$ 0.74	3.73 $\pm$ 0.55
Our approach	4.17 $\pm$ 0.50	4.07 $\pm$ 0.54	3.97 $\pm$ 0.56	4.13 $\pm$ 0.48

Table 1: Average User Rating on a scale of 1-5 for each dimension, contrasting the performance of our approach with the Attribute Conditioned LayoutGAN based approach

those generated using the LayoutGAN approach demonstrating the efficacy of our content-aware harmonization approach.

To further compare design layouts, we asked each worker to judge (i) if the text elements occupied “uncluttered” regions of the background image, and (ii) if the text appeared aligned. Banners generated using LayoutGAN almost always placed the text elements in the center of the background image. This is reflected in the poor saliency score of 2.92 for these set of banners. The saliency score of 4.07 for our approach reaffirms the fact that the placements were indeed context-aware. Since the LayoutGAN also accounts for the alignment of different elements, we expectedly observe less difference between the alignment scores of the two approaches. Finally, color harmonization scores across both approaches indicate the efficacy of our color selection module, as the scores obtained indicate reasonable satisfaction.

Finally, annotators were asked if they would make significant changes to the auto-created design suggestions, enabling us to evaluate the efficiency of our method in performing human-like harmonization. On a scale of 1 – 5, 1 indicating minimal change and 5 indicating drastic change, 90% of our designs had a score of ≤ 3 . On the other hand, 45% of LayoutGAN-based designs had a ≥ 4 . This re-enforces that our pipeline can generate banner design variations which do not require significant alteration, outperforming the baseline approach.

6 CONCLUSION

In this paper, we have studied the problem of composing image and text elements into an aesthetically pleasing banner design. Our system generates a layout for the design assets in a context-aware manner via a genetic algorithm framework. We predict appropriate font style for the design by considering image semantics and automate the color choice for each text element by maximizing visibility while maintaining color harmony with the rest of the image. Viability of the proposed approach is demonstrated both quantitatively and through a crowdsourced experiment.

While we have considered layout, font style, and color as axes along which harmonization may be performed, future work can extend the treatment presented here to other stylistic components of design such as patterns or shapes. In addition, our approach does

not investigate stylistic changes that can be made to the images themselves, such as cropping, applying filters, or color and style transfers, and we leave this for future explorations. Quantitatively scoring the aesthetics of a generated design is also an interesting direction for future research.

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